

llmXive Automated Review of DeTA: Discriminative Token Credit Assignment for Reinforcement Learning from Verifiable Rewards

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

This review focuses on the accuracy of factual claims and their supporting citations.

1. Model Availability and Citation Validity (Fatal): The paper claims to train and evaluate on “Qwen3-8B-Base” and “Qwen3-14B-Base” (Abstract, Section 5). The bibliography references yang2025qwen3. However, “Qwen3” is not a widely recognized public model as of the current knowledge cutoff. The provided LaTeX source lacks the full .bib content, making it impossible to verify if yang2025qwen3 is a legitimate technical report. If the model is hypothetical, unreleased, or the citation is invalid, the core experimental claims are unsupported. This is a critical failure in claim accuracy.

2. Ambiguity in Baseline Definition (Writing): The Abstract and Section 5.1 state DelTA outperforms “strongest same-scale baselines” by 3.26 and 2.62 points. Table 1 lists a row “Best Baseline”. The arithmetic ($28.40 - 25.14 = 3.26$) is correct. However, the phrasing “baselines” (plural) with a single delta is ambiguous. It is unclear if “Best Baseline” represents the single best method among {DAPO, SAPO, FIPO} or an average. The text should clarify if the delta is against the single best (matching the table header) or an average.

3. Consistency: Claims regarding code generation (Table 3) and OOD generalization (Table 4) are internally consistent with the provided numbers. The token weight analysis aligns with the text. The primary issues are the unverifiable model citation and the ambiguous baseline definition.

Recommendation: The authors must provide the full bibliography to verify the “Qwen3” model source. If the model is not publicly documented, the results cannot be verified. Additionally, clarify the baseline comparison method in the text to match Table 1.

Action items.

- **[writing]** Clarify if ‘Best Baseline’ in Table 1 is the single best method or an average. The text says ‘strongest baselines’ (plural) but reports a single delta. Match the text to the table header ‘Best Baseline’ (singular) or specify the aggregation method.
- **[fatal]** Verify the citation for ‘Qwen3-8B-Base’ and ‘Qwen3-14B-Base’. The model ‘Qwen3’ is not a standard public release. Ensure the reference ‘yang2025qwen3’ in the bibliography actually documents this model, or replace with a verified public model to support the experimental claims.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 provides a clear, high-level schematic of the DelTA method, effectively visualizing the four-step process from token gradient analysis to weighted centroid refinement. The diagram is well-organized, and the symbols (e.g., $\bar{\mu}_+$, $\bar{\mu}_-$) are self-explanatory within the context of the visual flow, matching the overview provided in the caption.

Figure 2

The figure is incomplete and fails to match its caption, which promises three distinct plots (Reward, Response Length, Entropy) while only the Reward plot is visible.

Figure 3

Figure 3 is a line plot comparing training dynamics, but it is critically flawed by having no caption to explain the data or axes. Additionally, the legend lists ‘DAPO’ while the paper’s context implies a comparison involving ‘DelTA’, creating ambiguity about which method is actually being visualized.

Action items.

- **[fatal]** Figure 2: The caption describes a three-panel figure (‘Left: Reward; Middle: Response Length; Right: Entropy’), but the rendered image displays only a single ‘Reward’ plot. The ‘Response Length’ and ‘Entropy’ panels are missing.
- **[fatal]** Figure 3: The caption is explicitly ‘(no caption)’, providing no context for the plot’s content, axes, or experimental setup.
- **[science]** Figure 3: The legend includes ‘DAPO’, but the caption for Figure 2 describes the comparison as ‘DelTA compared with DAPO’, suggesting the plot is intended to show DelTA but is mislabeled or the caption is missing the necessary context to verify the method being plotted.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on specialized terminology and acronyms that are not defined upon first introduction, creating a barrier for non-specialist readers. In the Abstract and Introduction,

the term “RLVR” is used immediately without spelling out “Reinforcement Learning from Verifiable Rewards.” Similarly, “DAPO” and “FIPO” are introduced in the Experiments section and Preliminaries without definition, despite being critical baselines.

In Section 3.1, the phrase “side-wise centroids” is used. “Side-wise” is not standard mathematical or statistical terminology; “side-specific” or explicitly “positive/negative-side” would be clearer. The Appendix introduces “stop-gradient” as a noun (“stop-gradient assignment score”) without defining the concept, assuming the reader knows this specific backpropagation technique. Additionally, the term “layer-restricted proxy” in the Appendix is used to describe a scalability optimization but lacks a plain-language explanation of what “layer-restricted” entails in this context.

To improve accessibility, the authors should expand all acronyms (RLVR, DAPO, FIPO, SAPO) at their first occurrence and replace non-standard jargon like “side-wise” with clearer alternatives. Defining technical constraints like “stop-gradient” and “layer-restricted” in a brief parenthetical or footnote would also help broaden the paper’s audience.

Action items.

- **[writing]** The manuscript relies heavily on specialized terminology and acronyms that are not defined upon first introduction, creating a barrier for non-specialist readers. In the Abstract and Introduction, the term “RLVR” is used immediately without spelling out “Reinforcement Learning from Verifiable Rewards.” Similarly, “DAPO” and “FIPO” are introduced in the Experiments section and Preliminaries without definition, despite being critical baselines. In Section 3.1, the phrase “side-wise centroids” is u

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a coherent theoretical framework for DelTA, linking the discriminator view of RLVR updates to a token-level reweighting mechanism. However, several logical gaps exist between the stated premises and the final conclusions.

First, the central empirical claim in the Abstract and Introduction—that DelTA outperforms the “strongest same-scale baselines”—is not fully supported by the data presentation in Table 1. The table lists a generic “Best Baseline” row with aggregate scores but fails to explicitly name which specific baseline (SAPO, FIPO, or DAPO w/ FT) achieved these scores for each model size. Without this explicit mapping, the reader cannot logically verify that DelTA is indeed outperforming the *strongest* competitor, as the “Best Baseline” could theoretically be a weaker method if the strongest one performed poorly or was excluded. The text in Section 5 (Significance Test Details) mentions SAPO and FIPO as the strongest for 8B and 14B respectively, but this connection is not made in the main results table where the claim is made.

Second, the mathematical derivation in Section 3.2 contains a logical discontinuity. The authors

define an optimization problem (Eq 3) to estimate the discriminative score $\alpha_{i,t}$, which includes an entropy regularizer term $\gamma + \sum_k h(\alpha_k)$. They then immediately present a closed-form solution (Eq 4) as a sigmoid function of the distance margin. The text does not demonstrate the calculus or algebraic steps required to derive the sigmoid form from the specific entropy function $h(\alpha)$. While the result is plausible (resembling a logistic regression or softmax derivation), the omission of the derivation step breaks the logical chain from the defined objective to the proposed algorithm.

Finally, the analysis in Section 5 (Q2) makes a strong causal claim: “bottom-50% collapses.” This conclusion is used to argue that the learned coefficients capture essential signals. However, the text provides no quantitative evidence (e.g., a specific accuracy drop or a plot) to substantiate the word “collapses.” A logical review requires that such strong descriptive terms be backed by the data they describe; without the specific metric showing the performance degradation, the conclusion remains an unsupported assertion.

Action items.

- **[writing]** The paper presents a coherent theoretical framework for DelTA, linking the discriminator view of RLVR updates to a token-level reweighting mechanism. However, several logical gaps exist between the stated premises and the final conclusions. First, the central empirical claim in the Abstract and Introduction—that DelTA outperforms the “strongest same-scale baselines”—is not fully supported by the data presentation in Table 1. The table lists a generic “Best Baseline” row with aggregate scores but

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several strong claims regarding the generalization and consistency of DelTA’s performance that appear to exceed the statistical rigor presented in the supplementary experiments.

First, the Abstract and Introduction claim that DelTA “generalizes to other backbones” and “improves code generation.” While the Appendix provides average scores for Olmo3-7B and code benchmarks (HumanEval+, MBPP+, LCB), these sections lack the rigorous statistical analysis (e.g., Mann-Whitney U tests, variance across seeds) provided for the main math benchmarks. A single average improvement on code (47.7 to 49.5) without per-benchmark significance or variance bars makes the claim of “improvement” potentially over-optimistic if the gain is driven by a single dataset. The authors should either provide the full statistical breakdown for these secondary tasks or qualify the language to reflect that these are preliminary observations.

Second, the Conclusion states DelTA “consistently improves... out-of-domain tasks.” The OOD results (Appendix E.5) show gains on GPQA-D and MMLU-Pro. However, the gain on MMLU-Pro for the 14B model is 1.63 points (66.77 to 68.40). Without a significance test for this specific

comparison, labeling it as a “consistent” improvement across tasks is an over-interpretation of the data. The term “consistent” implies a robust, statistically significant trend, which is not demonstrated for the 14B OOD results.

Finally, the theoretical framing claims DelTA “reshapes the update direction” by amplifying specific gradient directions. This is derived from a “local discriminator view” using a proxy gradient $\nabla(1-p_t)$ and a single refinement iteration ($K=1$). The paper does not sufficiently discuss the gap between this proxy-based approximation and the true gradient geometry. Claiming to reshape the *actual* update direction based on a proxy and a single step is a theoretical overreach. The limitations section should explicitly state that the “discriminator view” is an approximation and that the true gradient alignment may differ from the proxy-based analysis.

Action items.

- **[science]** Claims of ‘generalization’ to code/Olmo3 lack statistical significance tests or per-benchmark variance, unlike main math results. Provide full stats or temper claims to ‘preliminary promise’.
- **[writing]** Conclusion claims ‘consistent’ OOD improvement, but 14B MMLU-Pro gain (1.63 pts) lacks significance testing. Clarify ‘consistent’ scope or add per-task significance tests.
- **[writing]** Theoretical claim of ‘reshaping update direction’ relies on a proxy gradient and $K=1$ step. Explicitly acknowledge this approximation in limitations to avoid over-claiming true gradient geometry.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript addresses safety and ethics primarily through a brief “Broader impacts” appendix, which is insufficient for a method designed to optimize reasoning models. While the authors acknowledge the potential for misuse in “harmful code/content generation,” the discussion lacks specificity regarding the unique risks introduced by DelTA’s mechanism.

Specifically, the method amplifies token gradients that are discriminative between positive and negative advantage sides. In a safety context, this could inadvertently amplify “jailbreak” patterns or adversarial reasoning chains if the reward function (even a verifiable one like math correctness) is imperfectly aligned or if the “negative” side contains subtle safety violations that are not penalized. The review requires a more rigorous analysis of how DelTA might interact with safety filters or amplify biases present in the “DeepMath-103K” dataset.

Regarding data ethics, the paper relies on “DeepMath-103K” and benchmarks like AIME and HMMT. The authors must explicitly state the data provenance. If the dataset was scraped from the web, there must be a statement confirming that it does not contain Personally Identifiable Information (PII) and that the usage complies with the terms of service of the source platforms.

The current text assumes the dataset is benign without justification.

Finally, the “Broader impacts” section should be expanded to include a concrete mitigation strategy. For instance, if DelTA is used to train a model for code generation, how does the authors’ framework prevent the model from learning to generate malicious code that happens to pass a functional test? A generic call for “responsible deployment” is not a sufficient safety control for a high-impact RL method.

Action items.

- **[writing]** The ‘Broader impacts’ section (Appendix) is too generic. Explicitly detail specific dual-use risks (e.g., generating adversarial code, bypassing safety filters in reasoning models) and propose concrete mitigation strategies beyond ‘responsible deployment’.
- **[writing]** The paper uses ‘DeepMath-103K’ and benchmarks like AIME/HMMT. Clarify the data provenance and consent status. If scraped from the web, confirm compliance with terms of service and absence of PII or copyrighted material in the training set.
- **[writing]** The method reweights token gradients to amplify ‘discriminative’ signals. Discuss potential for ‘reward hacking’ or amplifying subtle biases in the reward function (e.g., favoring specific reasoning styles that correlate with demographic biases in the training data).

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The scientific evidence supporting the central claims of DelTA is generally robust, particularly regarding the consistency of improvements across multiple model scales (8B, 14B, 7B) and domains (math, code, OOD). The use of a large dataset (DeepMath-103K) and multiple benchmarks (7 math tasks) provides a solid foundation for the primary claim of improved token credit assignment. The theoretical derivation linking the update direction to a discriminative signal is logically sound and well-motivated.

However, the statistical rigor in reporting the primary results requires strengthening. Table 1 presents the main performance gains (3.26 and 2.62 points) as single scalar values. While Appendix A notes that significance tests were performed using 16 evaluation runs, the main text fails to report the standard deviation or confidence intervals for these averages. In reinforcement learning, where performance can be highly sensitive to random seeds, reporting variance is essential to determine if the observed gains are robust or potentially artifacts of a specific seed. Without this, the magnitude of the improvement relative to the noise floor remains ambiguous.

Additionally, the ablation study in Section 5 (Analysis) makes a strong qualitative claim that training on the bottom-50% of tokens causes the model to “collapse.” The text asserts this outcome but does not provide the specific numerical performance drop (e.g., accuracy falling from 25% to 5%) or a p-value to substantiate the severity of this collapse. This omission weakens

the evidence for the necessity of the discriminative weighting mechanism.

Finally, the hyperparameter sensitivity analysis (Table 4) indicates that increasing the refinement depth K beyond 1 yields diminishing returns (23.27 vs ~ 22.2). While the authors attribute the choice of $K=1$ to computational efficiency, the small performance gap suggests the method is not highly sensitive to this parameter. The paper should explicitly state whether the difference between $K=1$ and $K=2$ is statistically significant, as this would clarify if the overhead is a necessary trade-off for marginal gains or if a simpler configuration ($K=0$ or $K=1$) is sufficient.

Action items.

- **[science]** The main results table (Table 1) reports single-point averages (28.40 vs 25.14) without standard deviations or confidence intervals. While Appendix A mentions 16 evaluation runs, the primary results section must report the variance (e.g., mean $\pm$ std) to assess the stability of the 3.26 point gain against random seed noise.
- **[science]** The claim that ‘bottom-50% tokens collapse’ (Section 5, Q2) lacks quantitative evidence. The text states the result but does not provide the specific performance metric (e.g., accuracy drop from $X\%$ to $Y\%$) or statistical significance for this ablation, making the magnitude of the effect unverifiable.
- **[science]** The hyperparameter sensitivity study (Table 4) shows the base configuration ($K=1$) outperforms $K=2$ and $K=3$, but the performance gap is small (23.27 vs 22.15/22.20). The paper should clarify if this difference is statistically significant or if the choice of $K=1$ is primarily driven by the 10.2% computational overhead rather than a large performance delta.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical rigor of the empirical evaluation in the DelTA manuscript is currently insufficient to fully support the claims of robust improvement. While the authors perform a significance test (Section 5.1, Appendix **app:sig**), the reporting is incomplete. The text states that a one-sided Mann-Whitney U test was used with $S=16$ evaluation runs and $p < 0.05$, but it omits the actual test statistics (U-value), the exact p-values, and any measure of effect size (such as rank-biserial correlation). In statistical reporting, stating only that $p < 0.05$ without the magnitude of the effect or the specific test statistic prevents readers from assessing the practical significance of the results versus mere statistical significance.

Furthermore, the primary results presented in Table 1 (Section 5.1) and the supplementary tables (e.g., Table 3 for Olmo, Table 5 for Code) report mean scores with two decimal places but completely lack measures of dispersion, such as standard deviations, standard errors, or confidence intervals. Reinforcement Learning experiments, particularly those involving stochastic

sampling and evaluation, inherently possess variance. Without reporting the variance across the 16 evaluation runs (or the training seeds if applicable), it is impossible to determine if the reported average improvements (e.g., +3.26 on Qwen3-8B) are statistically distinguishable from the noise of the evaluation protocol. The precision of the reported means (e.g., 28.40) creates a false sense of certainty when the underlying distribution is unknown.

Finally, the hyperparameter sensitivity analysis in Appendix [app:hyp-sensitivity](#) (Table 4) presents point estimates for different configurations (\$K=1, 2, 3\$) but does not provide statistical evidence that the performance drop for \$K>1\$ is significant. The claim that \$K=1\$ is “optimal” relies on visual inspection of the table rather than a formal comparison of the means with associated uncertainty. To meet the standards of statistical analysis for this venue, the authors must supplement their tables with standard deviations or confidence intervals and report full statistical test results (including effect sizes) for all comparative claims.

Action items.

- **[science]** Section 5.1 (Significance Test Details) reports a Mann-Whitney U test on $S=16$ evaluation runs but fails to report the U-statistic, exact p-values, or effect sizes (e.g., rank-biserial correlation). Without these metrics, the claim of ‘significant’ improvement ($p < 0.05$) cannot be independently verified or assessed for practical magnitude.
- **[science]** The main results (Table 1) report average scores across seven benchmarks with high precision (e.g., 28.40, 39.91) but omit standard deviations or confidence intervals. Given the stochastic nature of RL training and evaluation, the absence of variance estimates makes it impossible to determine if the reported gains (3.26 and 2.62 points) are robust or within the noise floor of the evaluation protocol.
- **[science]** The hyperparameter sensitivity study (Table 4) compares Base DelTA against variants ($K=2, K=3$) using point estimates only. No statistical test or error bars are provided to confirm that the observed performance drops for $K>1$ are statistically significant rather than random fluctuations, weakening the claim that $K=1$ is ‘optimal’.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript demonstrates a generally high standard of academic writing, with a clear logical flow from the problem statement to the proposed method and experimental validation. The abstract effectively summarizes the core contribution and results. However, several specific issues regarding clarity and definition require attention before the paper is considered final.

First, in Section 2 (Preliminaries), the mathematical notation for the DAPO surrogate objective contains a potential ambiguity. The denominator $\sum |o_i|$ is used, but it is not explicitly defined whether this sum represents the total number of tokens in the group or a

specific normalization constant. While context suggests the former, explicit clarification would prevent reader confusion.

Second, and more critically, Section 3.2 introduces the optimization objective for the discriminative score $\alpha_{i,t}^{(k)}$. The equation includes a term $\gamma_{+}^{(k)}h(\alpha)$, yet the function $h(\alpha)$ is never defined in the main text or the appendices. This omission renders the equation incomplete and the method difficult to reproduce. The authors must define $h(\alpha)$ (e.g., as an entropy regularizer or a specific penalty function) or remove the undefined term if it was a placeholder.

Third, while the results in Section 4.1 are clear, the sentence “DelTA improves average scores by 3.26 (8B) and 2.62 (14B)” is slightly informal. A more precise phrasing, such as “DelTA improves average scores by 3.26 points on the 8B model and 2.62 points on the 14B model,” would enhance readability and professional tone.

Finally, in the Appendix under “Token weight analysis,” the figure caption refers to “Token clouds,” while the body text uses “high- and low-weight token clouds.” Maintaining consistent terminology throughout the document will improve cohesion. Addressing these points will significantly enhance the clarity and precision of the manuscript.

Action items.

- **[writing]** In Section 2 (Preliminaries), the formula for J_{DAPO} uses $\sum |o_i|$ in the denominator. This notation is ambiguous; clarify if this represents the total token count across the group or a specific normalization factor to ensure mathematical precision.
- **[writing]** In Section 3.2, the definition of $\alpha_{i,t}^{(k)}$ includes a term $\gamma_{+}^{(k)}h(\alpha)$. The function $h(\alpha)$ is not defined in the text or the appendix, making the equation impossible to verify or reproduce.
- **[writing]** In Section 4.1 (Main Results), the text states ‘DelTA improves average scores by 3.26 (8B) and 2.62 (14B).’ These numbers match the difference in Table 1, but the sentence structure is slightly clunky. Consider: ‘DelTA improves average scores by 3.26 points on the 8B model and 2.62 points on the 14B model.’ for better flow.
- **[writing]** In the Appendix (Token weight analysis), the caption for Figure 1 refers to ‘Token clouds,’ but the text describes them as ‘high- and low-weight token clouds.’ Ensure consistent terminology between the figure caption and the main text description.